화학특강 (2009년 2학기)
제4장

황현석

2009년 9월 28일



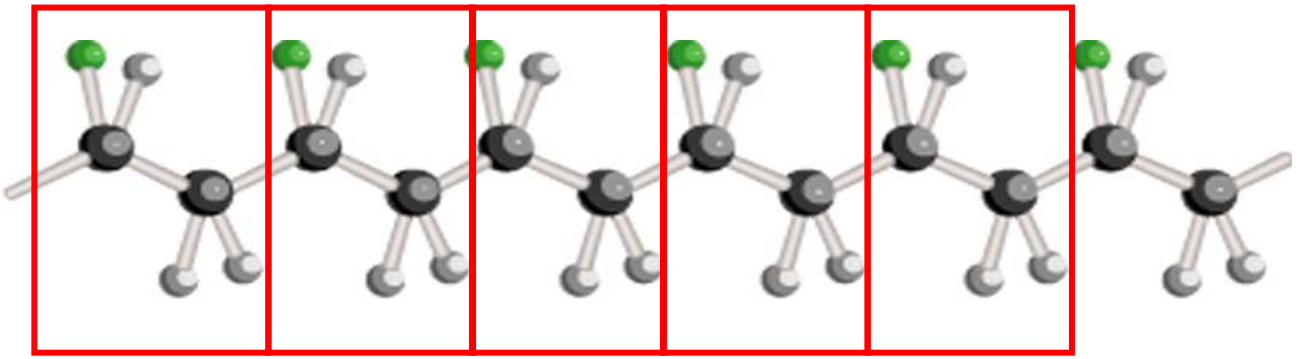
What is Nanotechnology?



Polymers

Greek: “polys” ~ many
“meros”~parts

Polymers are molecules consisting of a (large) number of structural units called monomers, that are connected by covalent bonds.



The monomers are small (organic) molecules, linked together in a process called polymerization.



ethylene — polyethylene

Chang, 25.1

Molar Mass Distributions of Polymers

- Degree of polymerization (N): the number of monomers in a polymer
- Monodisperse and monodisperse
  - If all polymers in a sample have the same number of monomers, the sample is **monodisperse**.
  - If polymers in a sample consist of a various number of monomers, the sample is **polydisperse**.
- Number-average molar mass:

$$M_n = \sum_N n_N M_N$$

- Weight-average molar mass:

$$M_w = \frac{\sum_N n_N M_N^2}{\sum_N n_N M_N} = \frac{\sum_N n_N M_N^2}{M_n}$$

- Polydispersity index

$$I_p = \frac{M_w}{M_n}$$

예제

- 몰질량이 10^5 g/mol인 gelatin monomer 단백질과 몰질량이 2×10^5 g/mol인 gelatin dimer 단백질을 포함하는 시료가 있다. gelatin monomer의 몰분율이 0.5라고 할 때, 시료 속 gelatin 단백질의 number-average 몰질량과 weight-average 몰질량을 구하여라. 이 단백질의 polydispersity는 얼마인가?

$$\text{number}-\text{average molar mass} = \sum_N n_N M_N = 1.5\times10^5 \text{ g/mol}$$

$$\text{weight}-\text{average molar mass} = \frac{\sum_N n_N M_N^{\,2}}{\sum_N n_N M_N} \cong 1.67\times10^5 \text{ g/mol}$$

$$I_p = \frac{M_w}{M_n} = \frac{10}{9}$$

Homopolymers and Copolymers

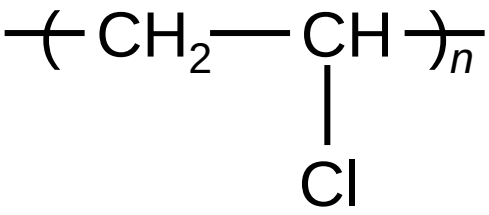
- ***Homopolymer*** is a polymer made up of only one type of monomer.



Teflon

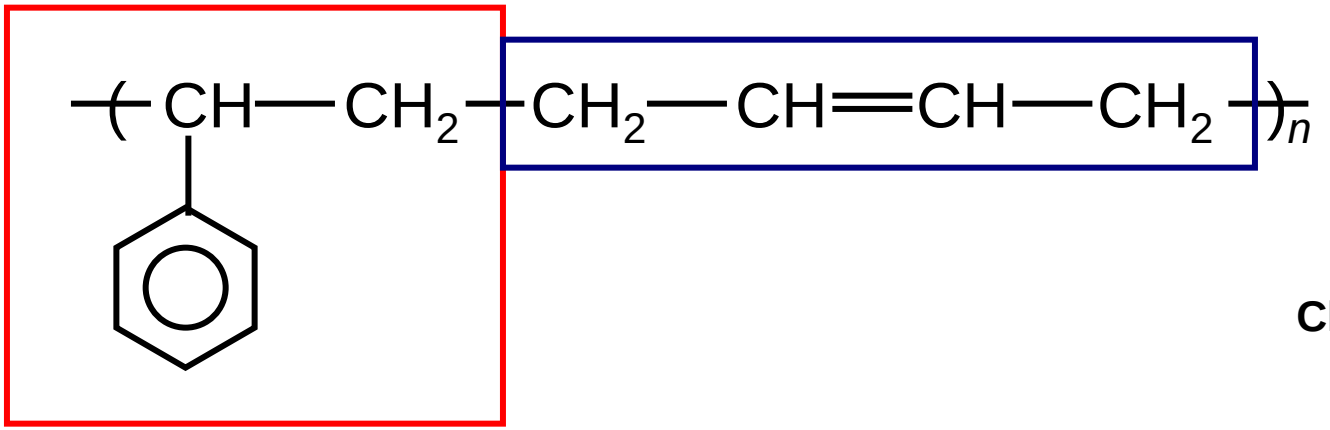


Polyethylene



PVC

- ***Copolymer*** is a polymer made up of two or more monomers.

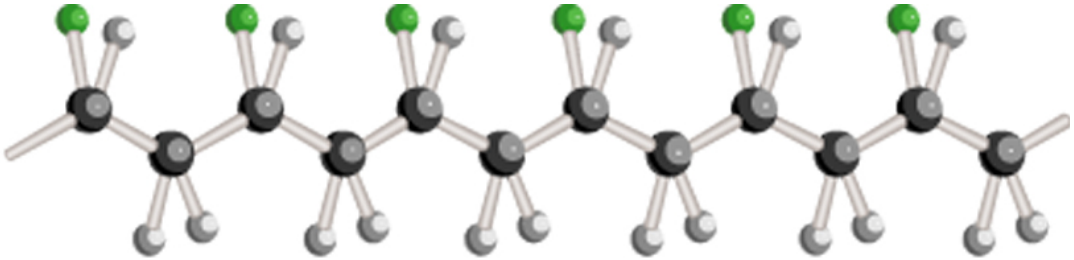


Chang, 25.1

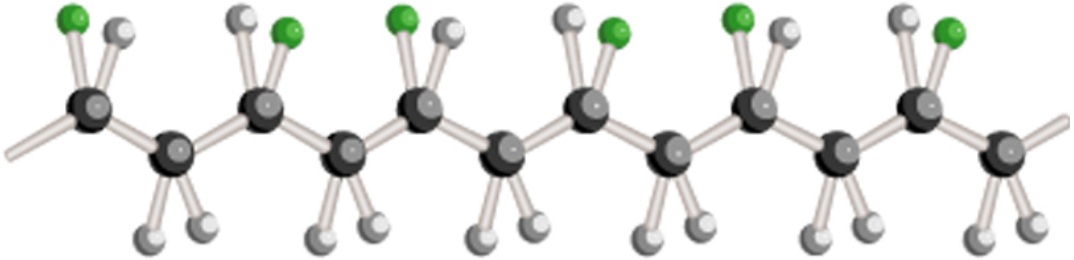
Styrene-butadiene rubber

Stereoisomers of Polymers

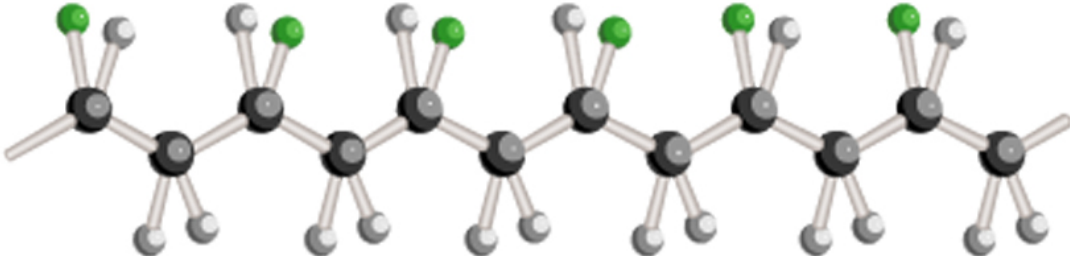
Isotactic: similar side groups always on same side of the chain. Highest melting point and great crystallinity



Syndiotactic: side groups on alternating sides of the chain

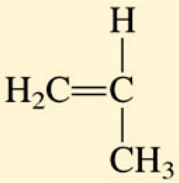
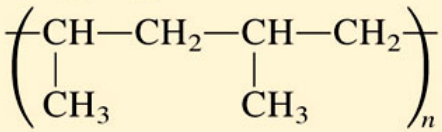
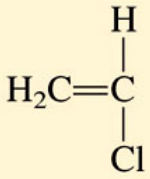
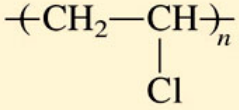
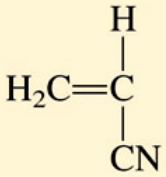
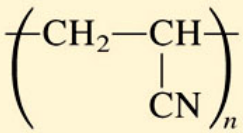


Atactic: side groups disposed at random. Rubbery,amorphous, and relatively weak



Common Polymers (1)

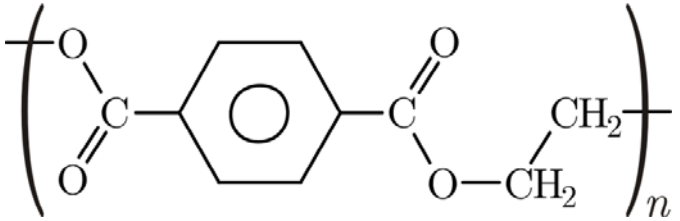
TABLE 25.1 Some Monomers and Their Common Synthetic Polymers

Monomer		Polymer	
Formula	**Name**	**Name and Formula**	**Uses**
$H_2C{=}CH_2$	Ethylene	Polyethylene $+CH_2{-}CH_2+_n$	Plastic piping, bottles, electrical insulation, toys
	Propylene	Polypropylene 	Packaging film, carpets, crates for soft-drink bottles, lab wares, toys
	Vinyl chloride	Poly(vinyl chloride) (PVC) 	Piping, siding, gutters, floor tile, clothing, toys
	Acrylonitrile	Polyacrylonitrile (PAN) 	Carpets, knitwear

Chang, 25.2

Common Polymers (2)

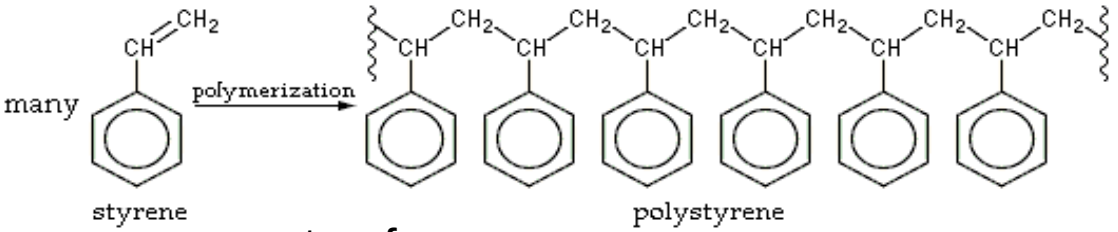
- Polyethylene terephthalate (PET)





- Polystyrene





styrofoam



- High-density polyethylene (HDPE) and low-density polyethylene (LDPE) and



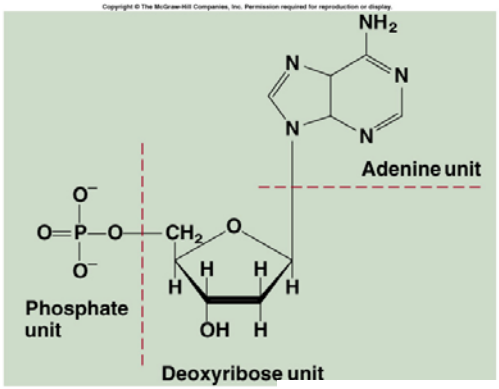




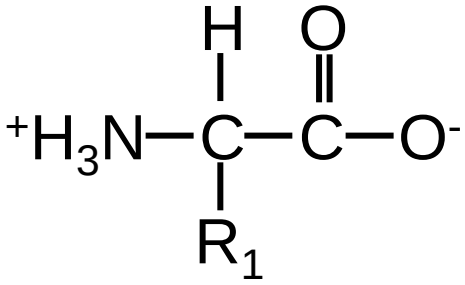


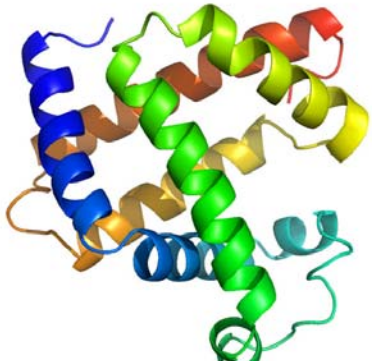
Biopolymers

- **DNA (RNA)** consists of subunits called nucleotides: sugar deoxyribose (ribose), phosphate group PO_4^- and a base.

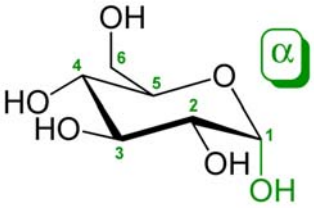


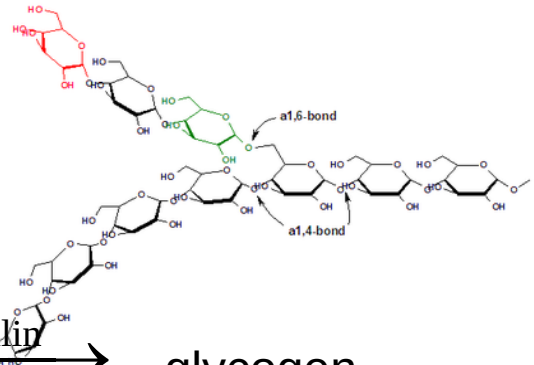
- **Proteins** are polypeptides with 20 different amino acids as the structural subunits.





- **Polysaccharides** are complex, often branched, carbohydrates, consisting of monosaccharides (simple sugars)





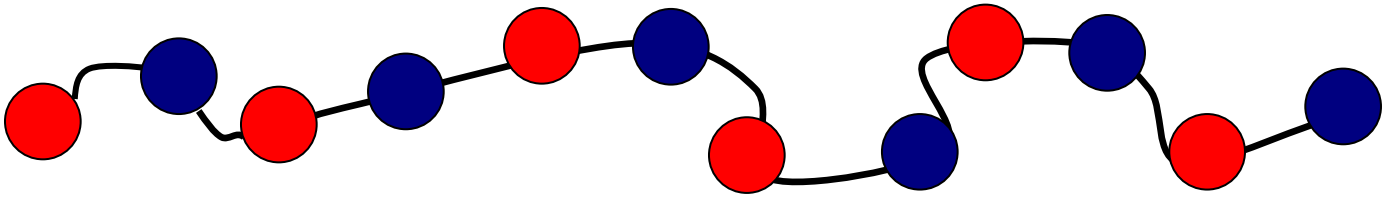
α-glucose ⇄ glycogen

insulin

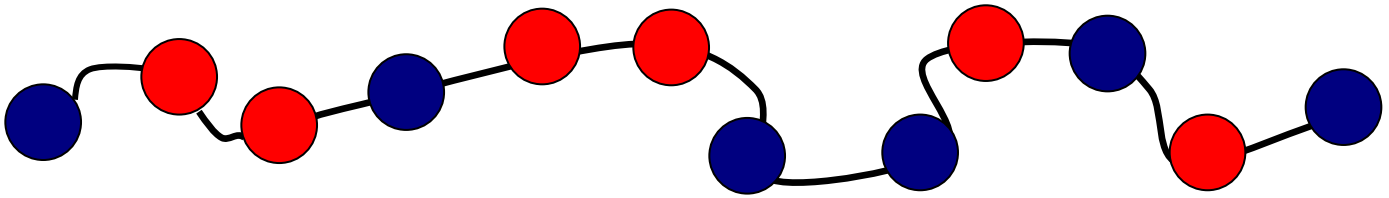
glucagon

Copolymers

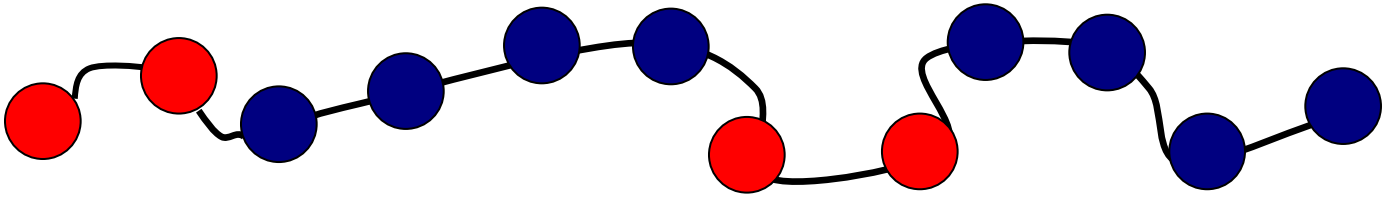
- Alternating copolymers



- Random copolymers

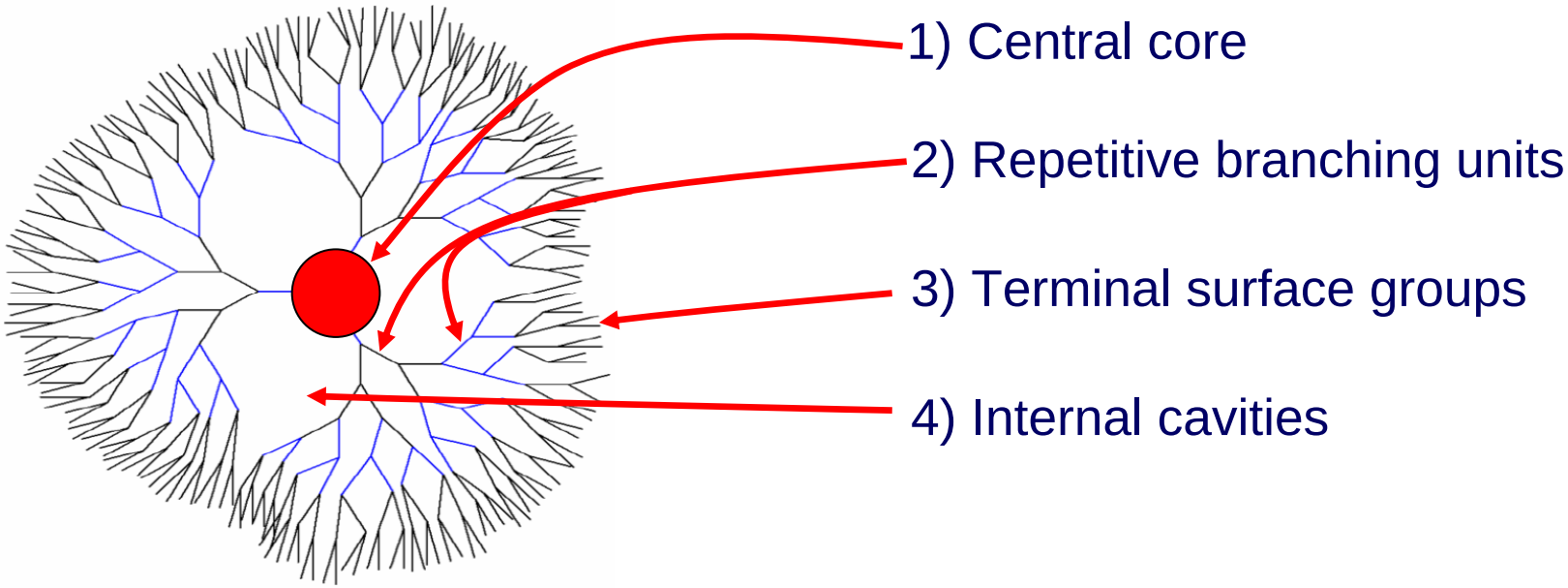


- Block copolymers



Dendrimers (1)

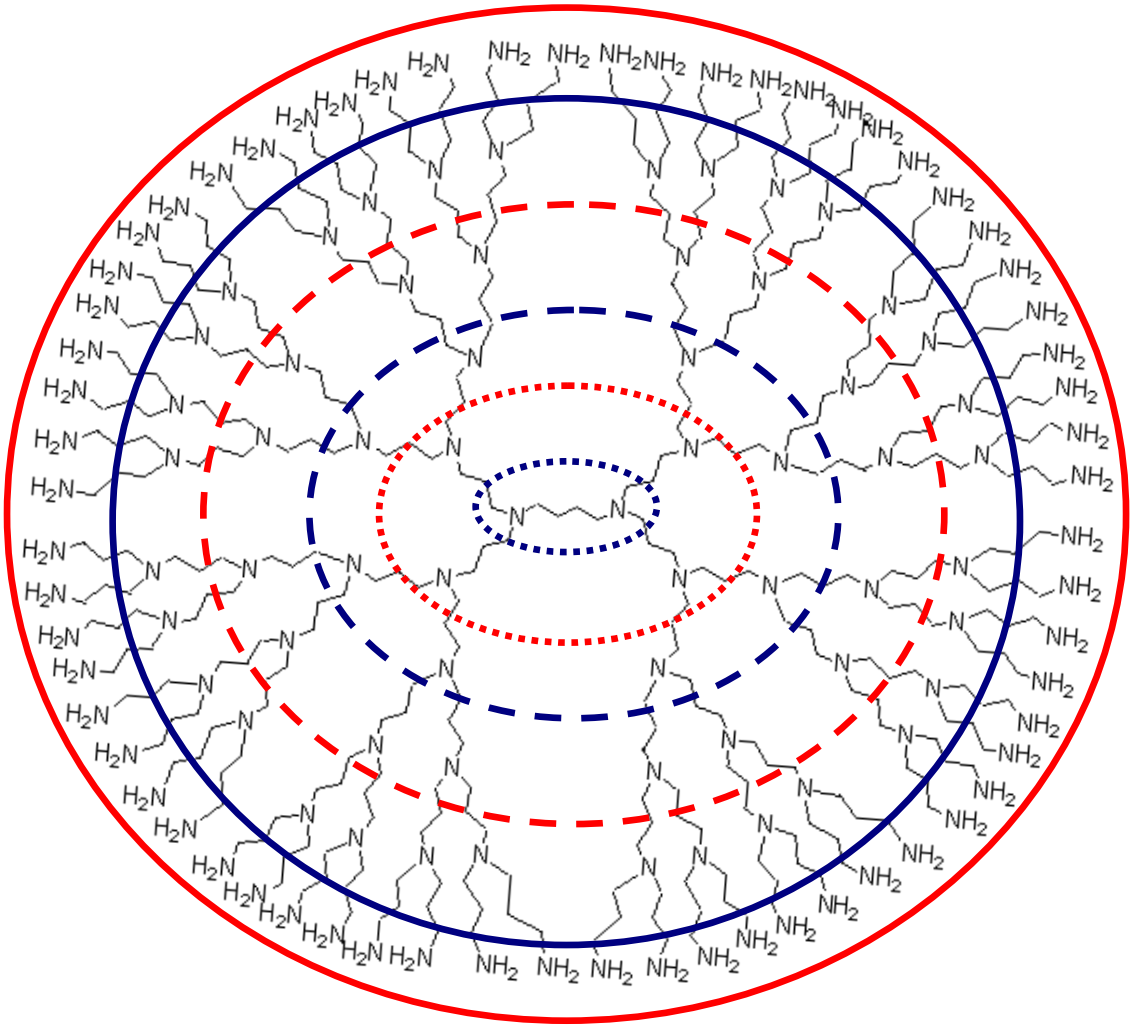
- Dendrimers: highly branched (hyperbranched) polymers that have a well defined structure. [“dendros” (Greek) ~ tree, branch



- Divergent synthesis: from the core to the surface
- Convergent synthesis: from the surface to the core

Dendrimers (2)

- Polyamidoamin (PAMAM) dendrimers



Central core moiety

1st generation

2nd generation

3rd generation

4th generation

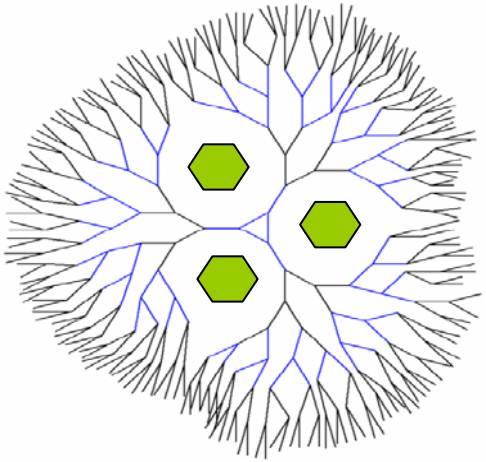
5th generation

Dendrimers (3)

- Properties of dendrimers are dominated by functional groups on the surface of the globular molecule. For example, a dendrimer can be made water-soluble with hydrophilic surface groups

  (e.g. R-COOH $\rightarrow$ R-COO$^-$ + H$^+$)

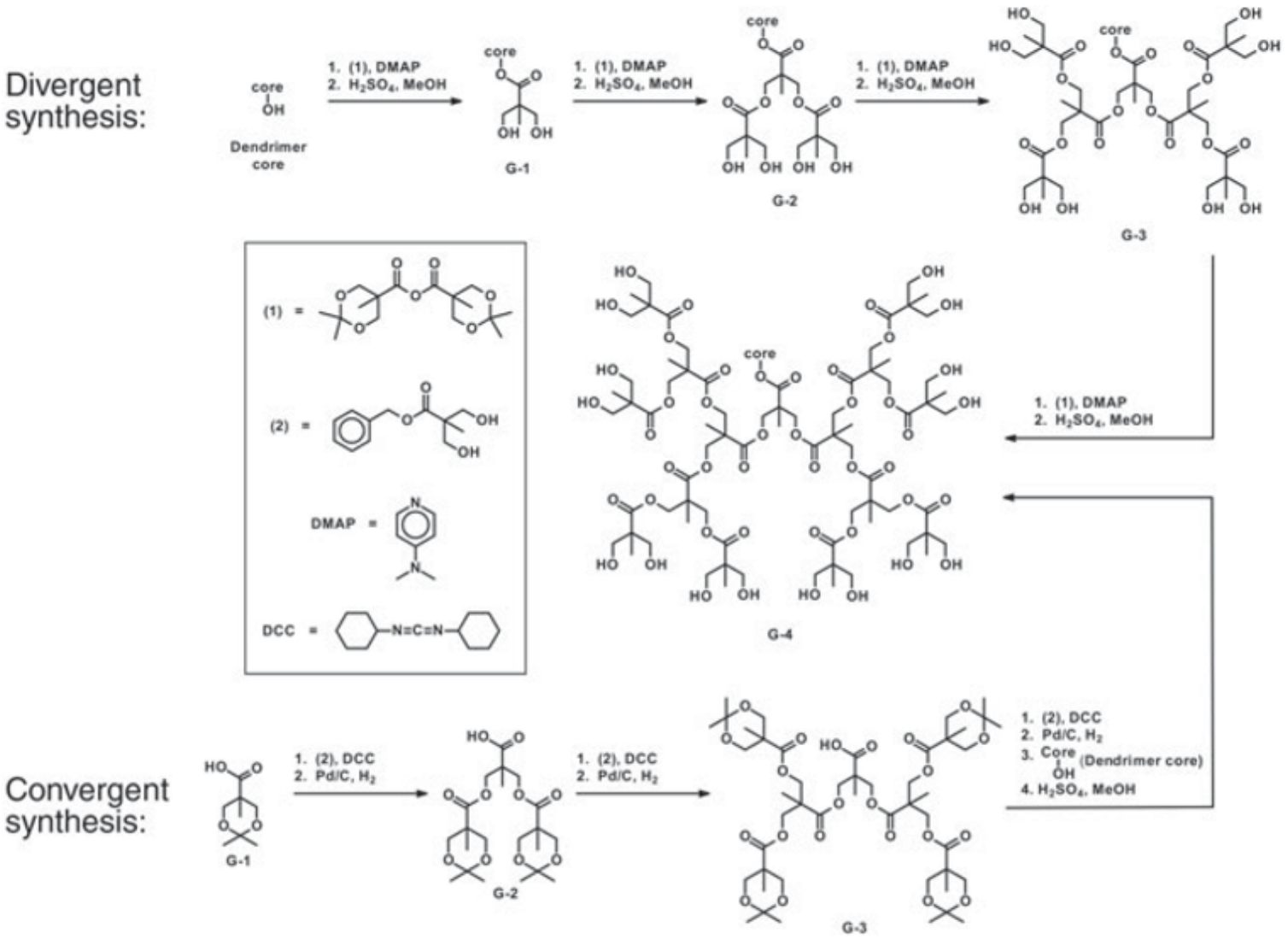
- Dendrimers with hydrophobic internal structure can act as drug carriers. Alternatively, the drug could be attached directly to the dendrimer surface groups.



= encapsulated drug

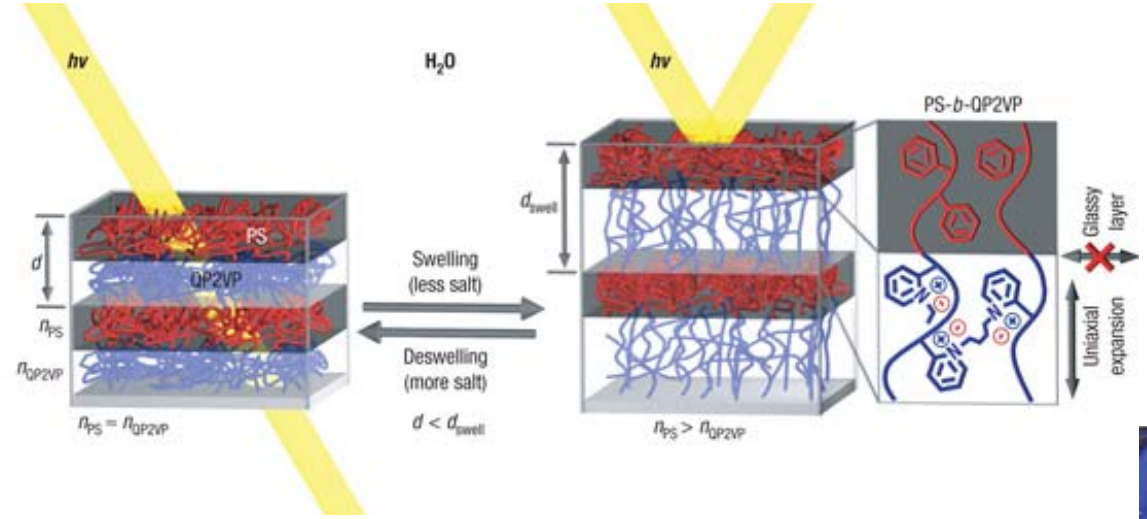
- Dendrimers have also shown promise in the development of organic LEDs: photons can be directly toward a luminescent core moiety by suitably chosen surface groups.

Dendrimer Synthesis via Divergent and Convergent Approaches

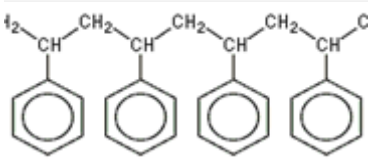


Lee et al., Nature Biotechnology **23**, 1517 (2005)

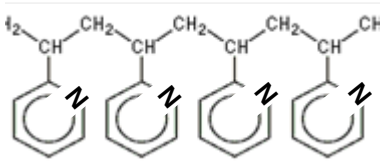
Block-Copolymer Photonic Gel







polystyrene

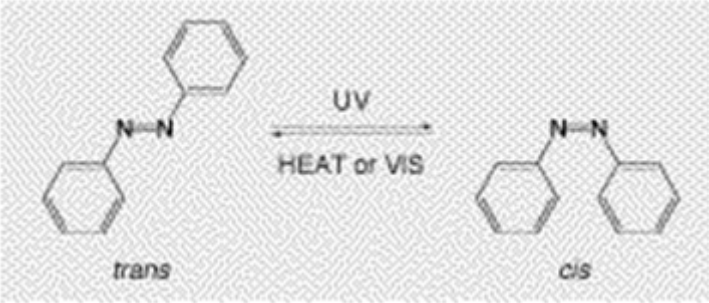


Poly(2-vinyl pyridine)

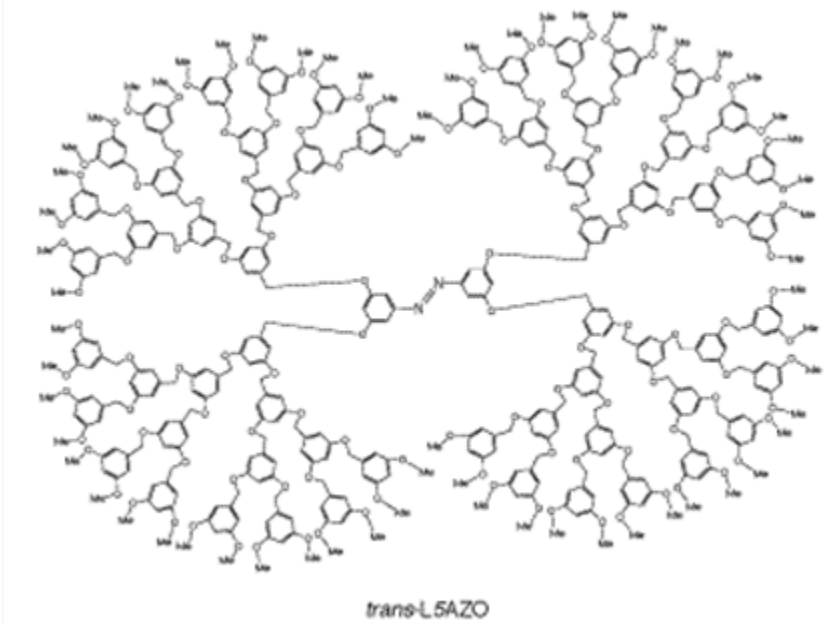
Kang et al., Nature Materials **6**, 957 (2007)

Photoisomerization in dendrimers by harvesting of low-energy photons

- Cis-trans isomerization of azobenzen



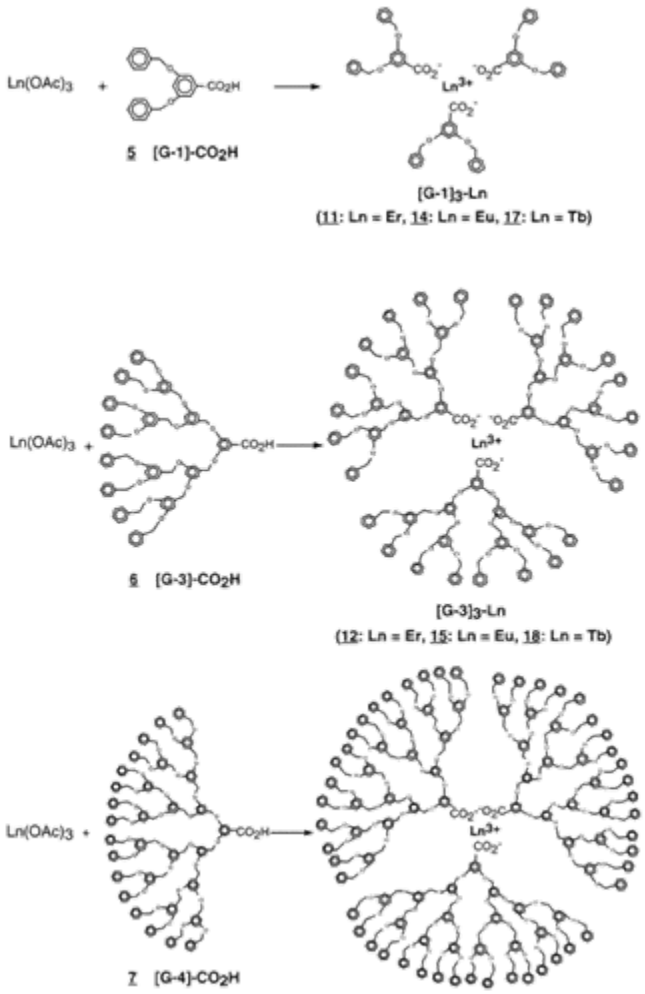
- Photoactive dendrimer system



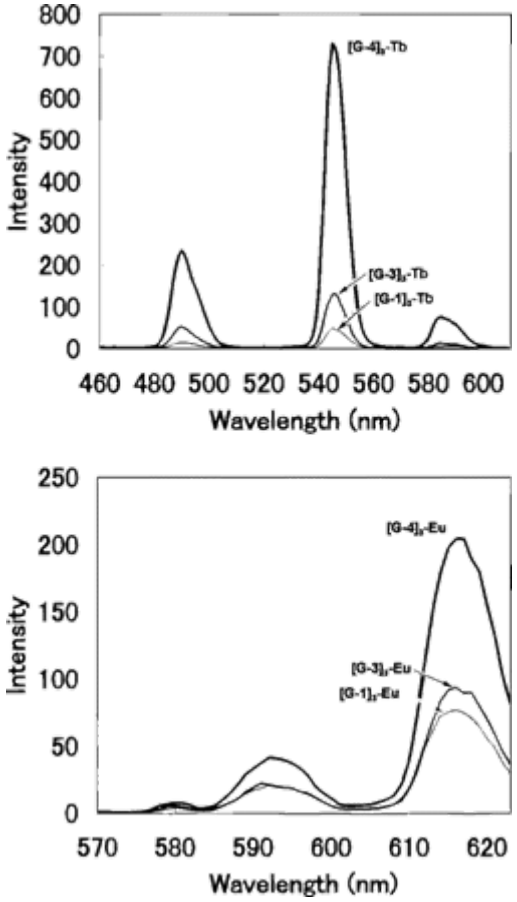
Fischer et al., Angew. Chem. Int. Ed. **38**, 884 (1999)

Luminescence Properties of Lanthanide Ions through Site-Isolation and Antenna Effects

- Ln-cored dendrimer complexes



- Emission spectra of Ln-cored dendrimer complexes



Kawa and Fréchet, Chemistry of Materials **10**, 286 (1998)

Colloids (1)

- Colloids: a dispersion of particles of one substance (the dispersed phase) throughout a dispersing medium maid of another substance.

TABLE 12.4 Types of Colloids

Dispersing Medium	Dispersed Phase	Name	Example
Gas	Liquid	Aerosol	Fog, mist
Gas	Solid	Aerosol	Smoke
Liquid	Gas	Foam	Whipped cream
Liquid	Liquid	Emulsion	Mayonnaise
Liquid	Solid	Sol	Milk of magnesia
Solid	Gas	Foam	Plastic foams
Solid	Liquid	Gel	Jelly, butter
Solid	Solid	Solid sol	Certain alloys (steel), gemstones (glass with dispersed metal)

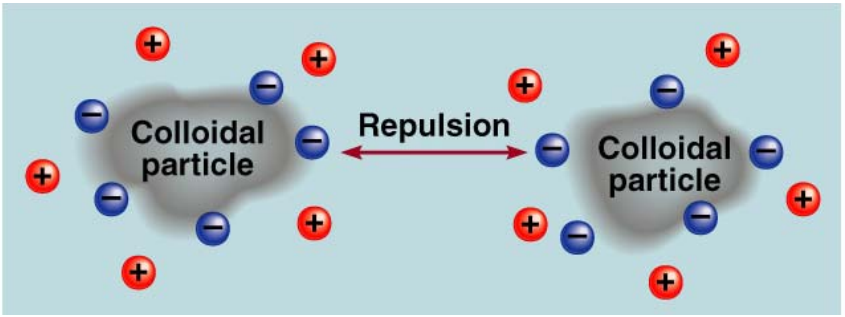
Chang, General Chemistry **12.8**

Colloids (2)

- Tyndall effect



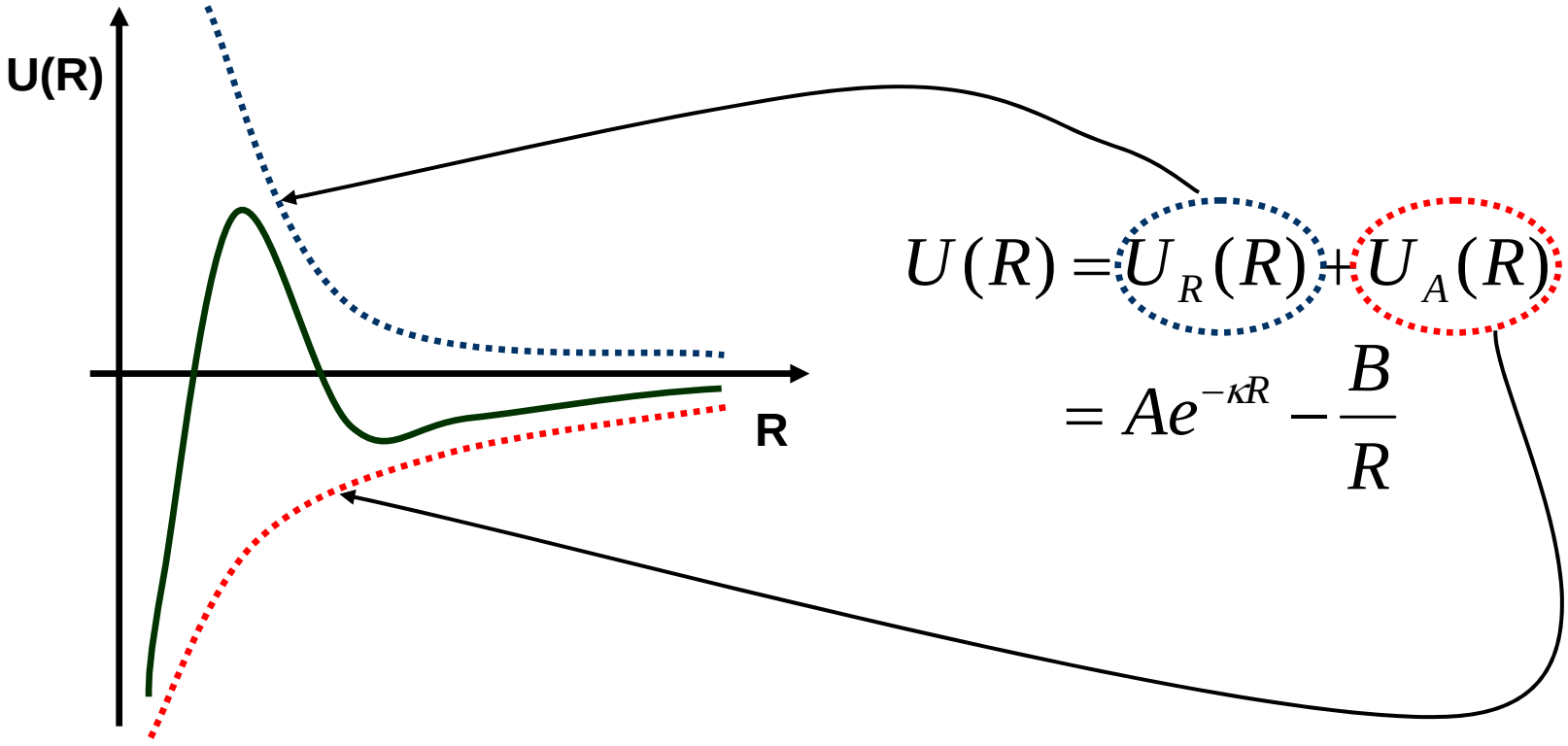
- Electrostatic repulsion



Chang, General Chemistry **12.8**

Colloids (3)

- Interactions between colloids



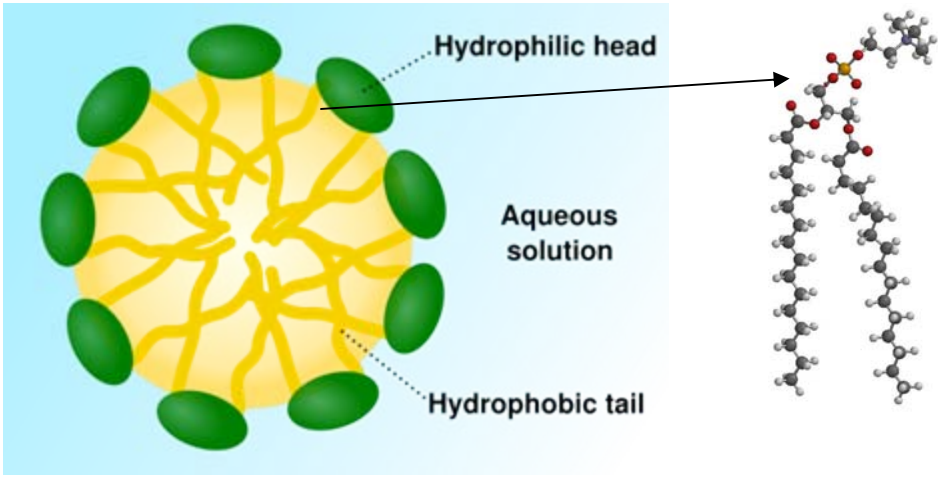
- Coagulation: the destruction of a colloid

Chang, General Chemistry **12.8**

Colloids (4)

- Mayonnaise





- River Delta

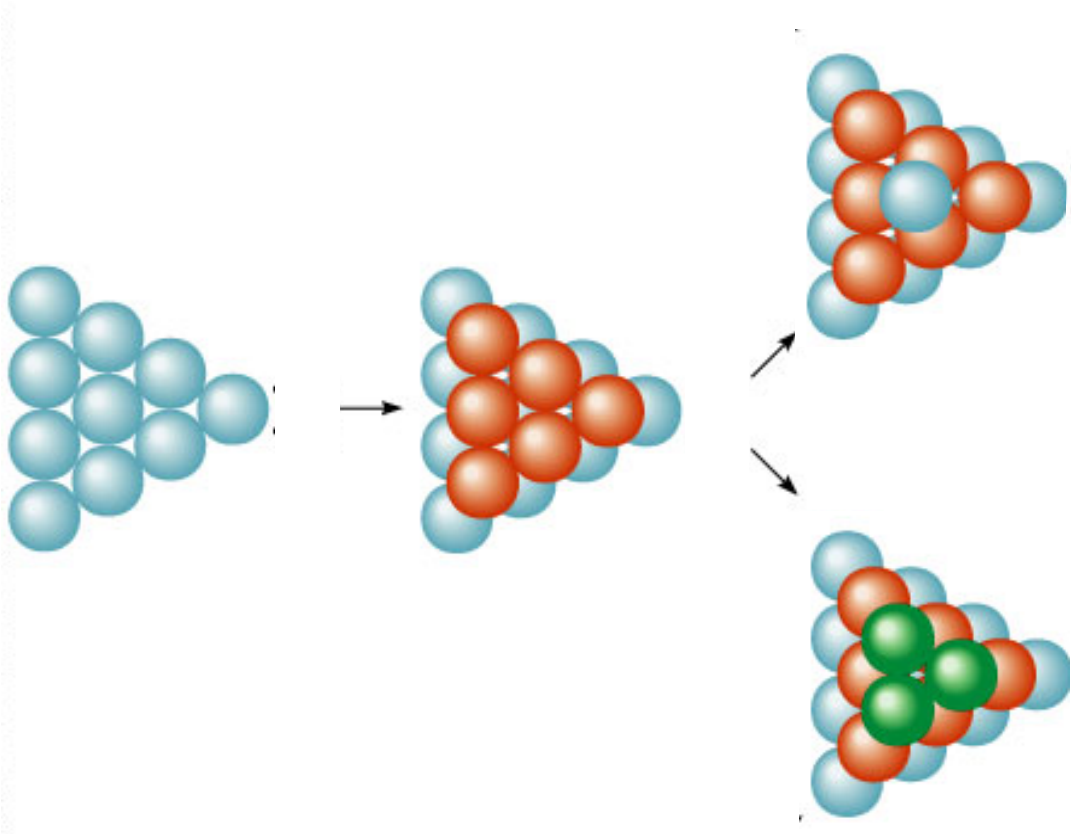


Nile Delta

Chang, General Chemistry **12.8**

Colloids and Crystallization

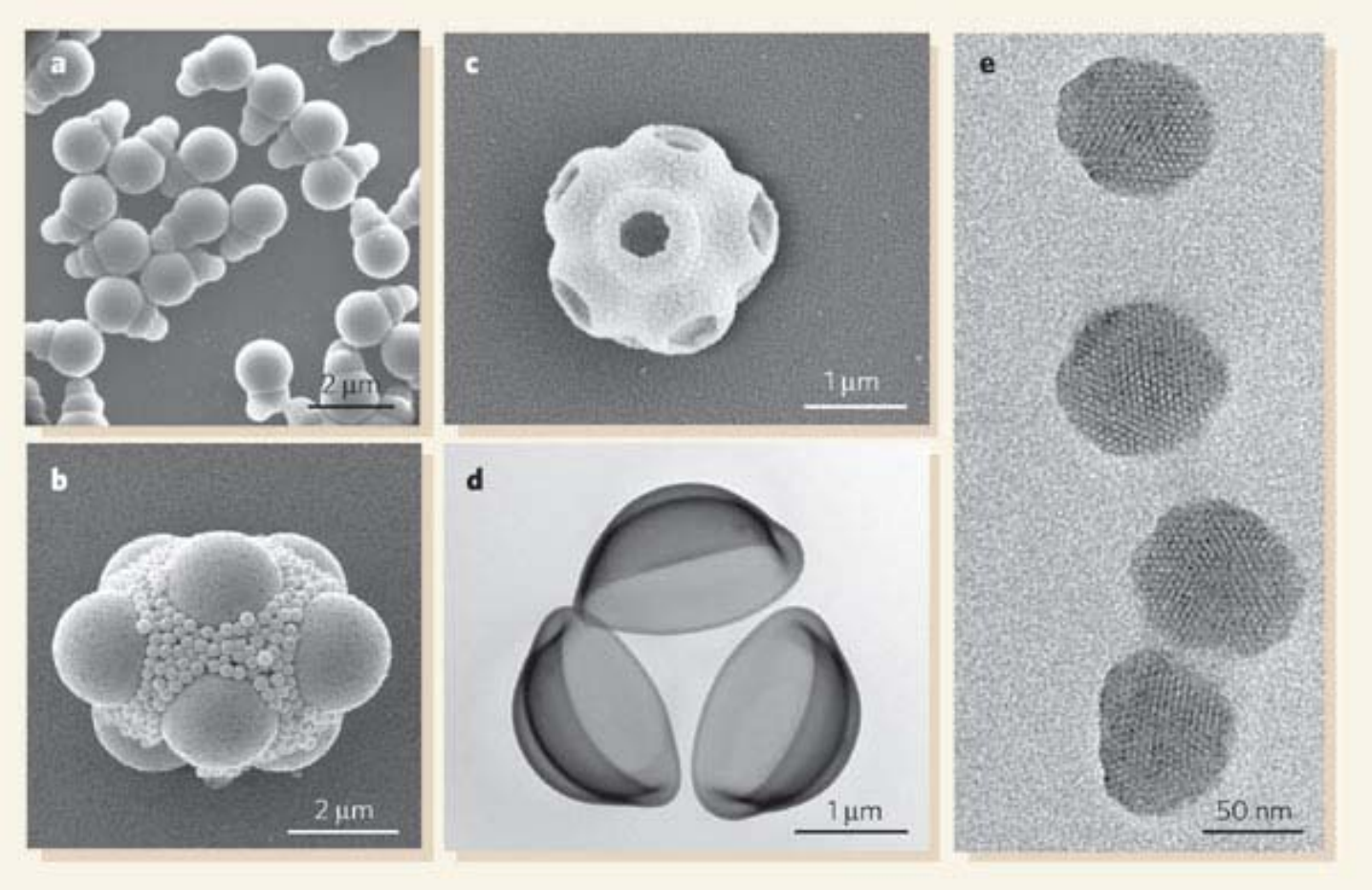
- Hexagonal and cubic closed packed structures



hexagonal closed packed structure (aba)

cubic closed packed structure (abc)

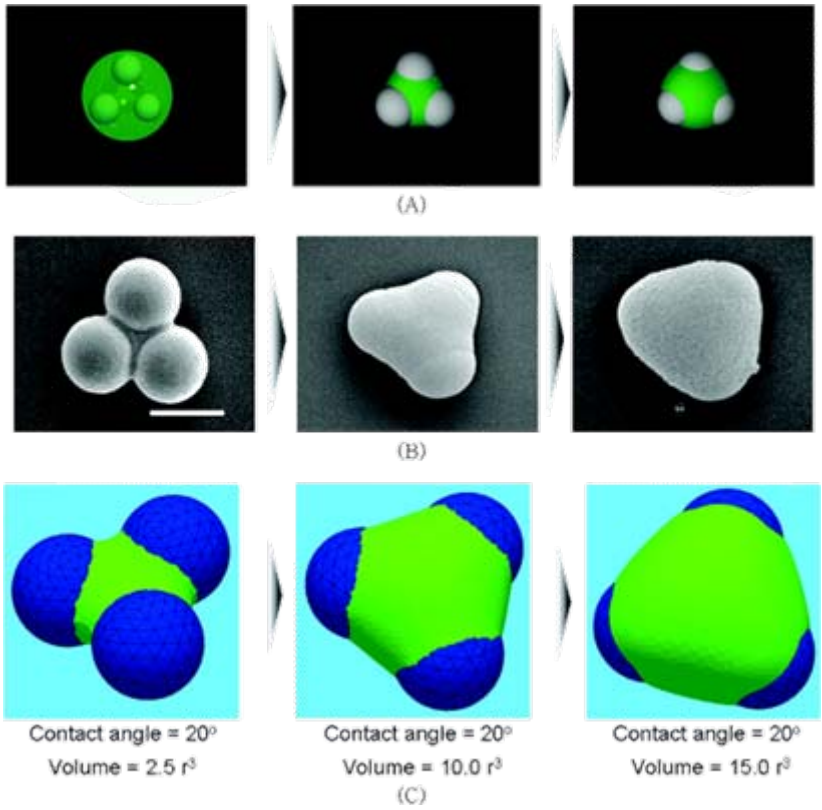
Colloids in the Nanotechnology

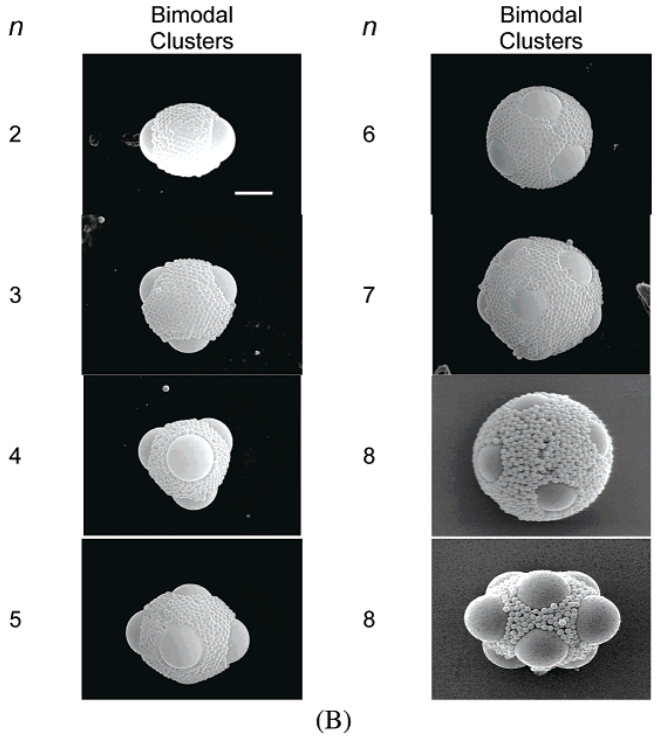


van Blaaderen, Nature **439**, 545 (2006)

Colloids in the Nanotechnology

- Formation of a trimer cluster of silica microparticles with silica nanoparticles from water-in-oil emulsions.

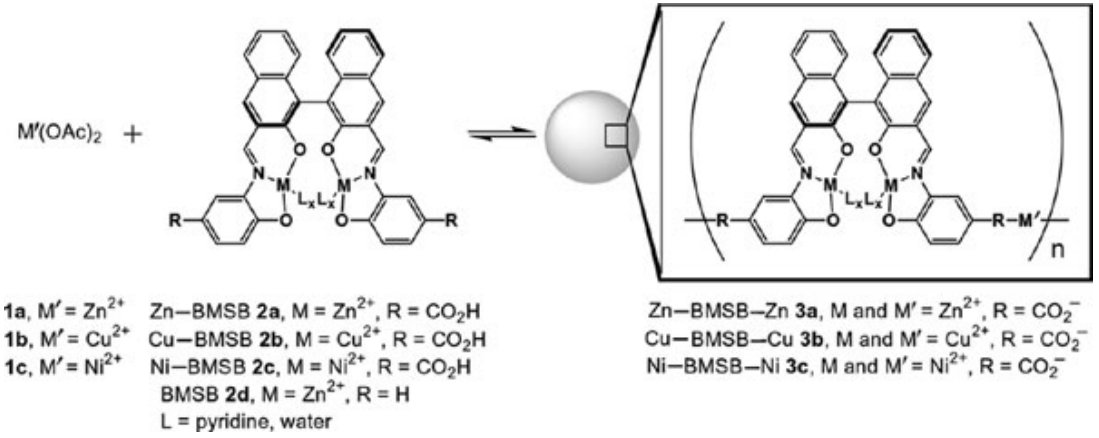




Young-Sang Cho, J. Am. Chem. Soc. **127**, 15968 (2005)

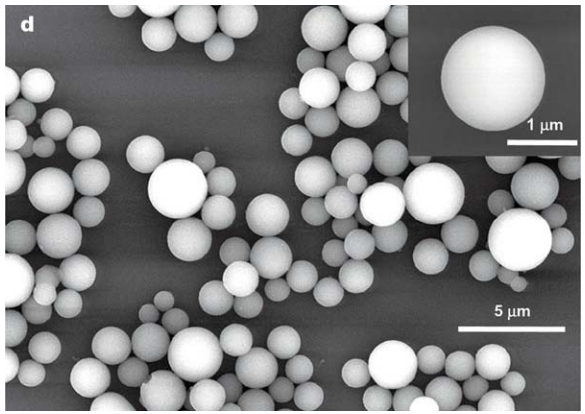
Colloids in the Nanotechnology

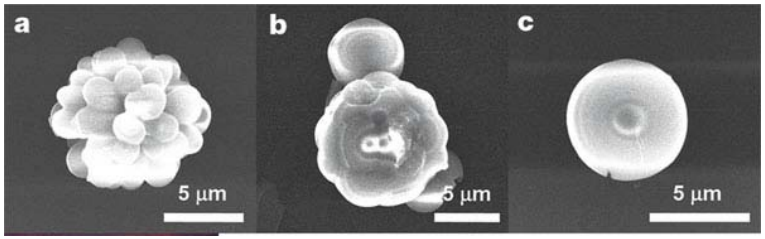
- assembly of metal ions and homochiral carboxylate-functionalized binapthyl bis-metallo-tridentate Schiff base (BMSB) building blocks

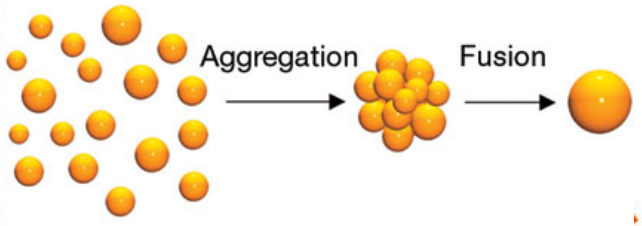


1a, M′ = Zn^{2+} Zn–BMSB **2a**, M = Zn^{2+}, R = CO_2H
1b, M′ = Cu^{2+} Cu–BMSB **2b**, M = Cu^{2+}, R = CO_2H
1c, M′ = Ni^{2+} Ni–BMSB **2c**, M = Ni^{2+}, R = CO_2H
BMSB **2d**, M = Zn^{2+}, R = H
L = pyridine, water

Zn–BMSB–Zn **3a**, M and M′ = Zn^{2+}, R = CO_2^-
Cu–BMSB–Cu **3b**, M and M′ = Cu^{2+}, R = CO_2^-
Ni–BMSB–Ni **3c**, M and M′ = Ni^{2+}, R = CO_2^-



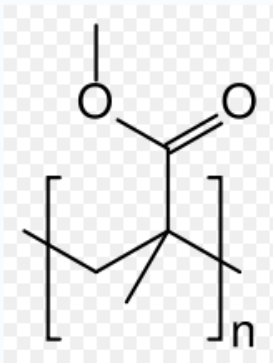




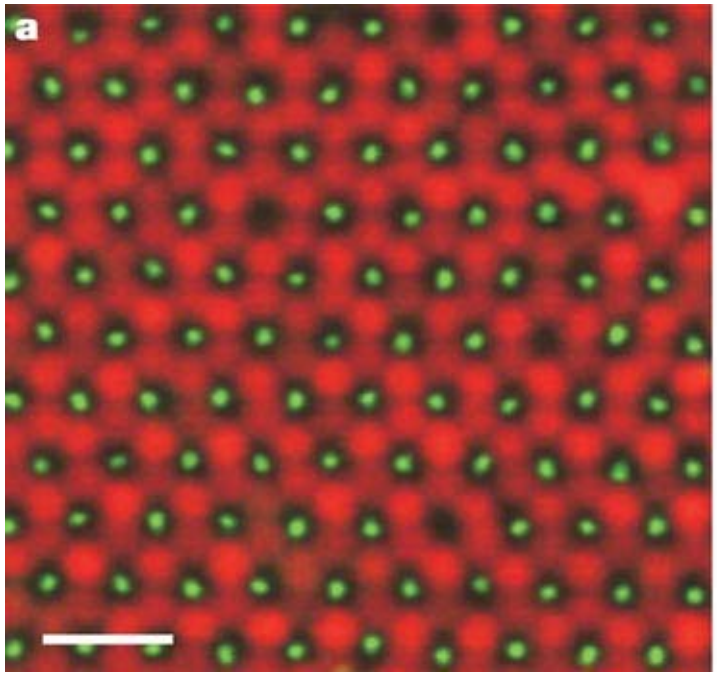
Oh and Mirkin, Nature **431**, 651 (2005)

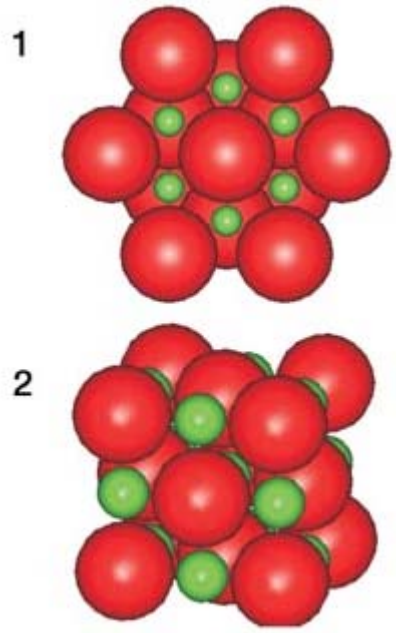
Colloids and Crystallization

- Poly(methyl methacrylate) (PMMA) colliods



- Crystallization of PMMA colloids





Leunissen, Nature **437**, 235 (2005)

The Nobel Prize in Chemistry 2008

"for the discovery and development of the green fluorescent protein, GFP"

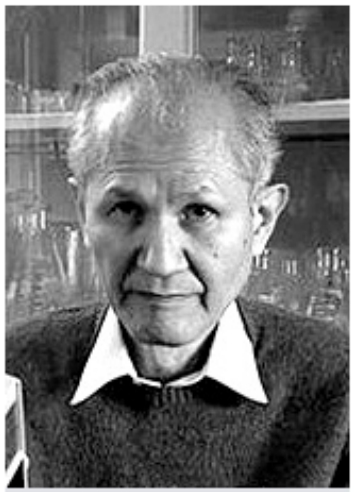


Photo: J. Henriksson/SCANPIX

Osamu Shimomura

1/3 of the prize

USA

Marine Biological Laboratory (MBL)
Woods Hole, MA, USA

b. 1928

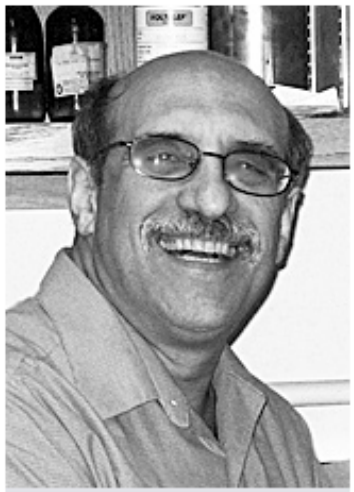


Photo: J. Henriksson/SCANPIX

Martin Chalfie

1/3 of the prize

USA

Columbia University
New York, NY, USA

b. 1947



Photo: UCSD

Roger Y. Tsien

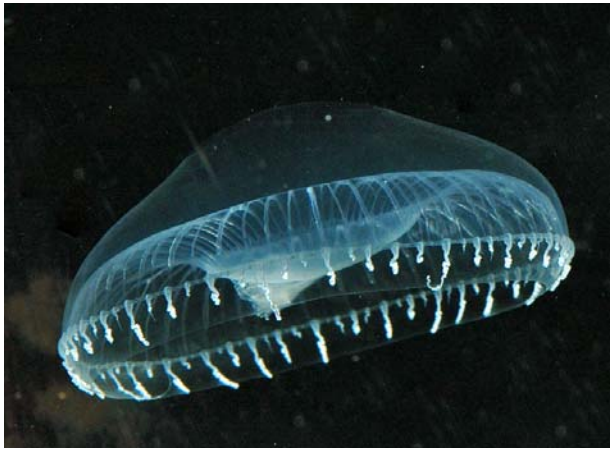
1/3 of the prize

USA

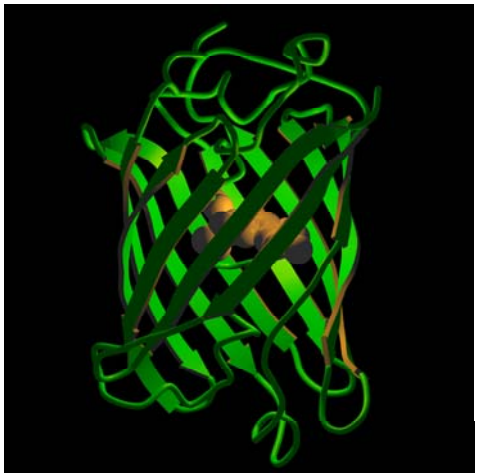
University of California
San Diego, CA, USA

b. 1952

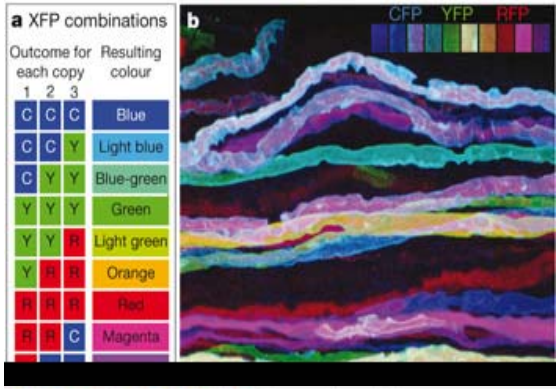
What is Green Fluorescent Protein (GFP)?



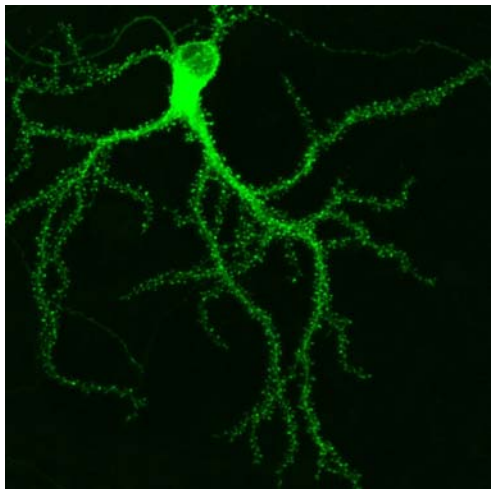
Aequorea victoria jellyfish



GFP



Livet et al., Nature **450**, 56 (2007)



Neuron GFP

